Reg. No. | | | | | | | | | | | [12EC51/13EC51]

FIFTH SEMESTER B.E/B.TECH EXAMINATION DECEMBER –2015

ANALOG COMMUNICATION

Time:3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions.
2. Missing data may be suitably assumed.

1. a. Consider the signal- $x(t)=e^{-at}.u(t),\ a>0$ where u(t) is the unit step function defined by $u(t) = \begin{cases} 1, & t > 0 \\ 0, & t < 0 \end{cases}$ **06**

Find the Fourier transform of x(t) and sketch its magnitude and phase spectra.

b. Find the Fourier transform of the signal $x(t)=\dfrac{\sin(at)}{\pi t}$. **04**

c. Let signal x(t) be real, and X(w)=F[x(t)] then show that, **04**

$f[x(-t)] = X(-w) = X(w)$

2. a. A message signal m(t) whose spectrum is as shown in Fig.2.2(a) is passed through the system Fig.2.1(a). The Band pass filter has a band width of 2W centered at f_o. Plot the spectra of the signal x(t), $y_1(t)$ and y(t). **08**

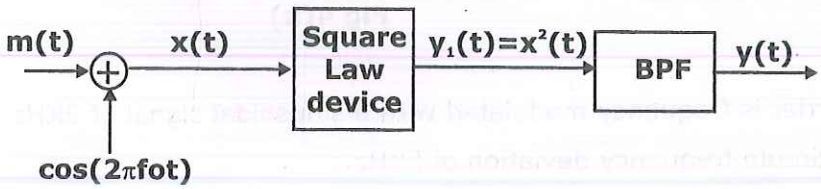

Fig 2.1(a)

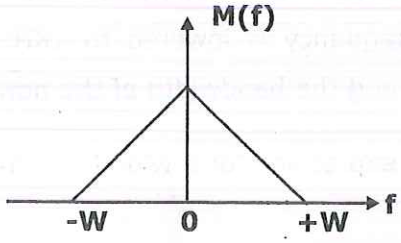

Fig 2.2(a)

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b. Explain the envelope detection of AM wave, using relevant waveform and equations. **06**

3. a. Explain how ring modular can be used to generate DSB-SC modulation with proper block diagram and necessary equations /expressions. **08**

b. A DSB modulated signal $\phi(t) = A.m(t).Cos(2\pi fct)$ is multiplied with a local carrier $c(t) = \cos(w_o t + \theta)$ and the output is passed through a LPF with a bandwidth equal to the band width of the message m(t). Determine. **06**

i. The power of modulated signal

ii. The power of the signal at the output of the LPF.

4. a. Explain the generation of SSB wave using phase discrimination method with the help of neat functional block diagram and relevant waveforms. **08**

b. The frequency response H(w) of a VSB filter is shown in Fig. 4(b). **06**

i. Find the USB signal $x_{usb}(t)$ *when* $m(t) = a_1 \cos(w_1 t) + a_2 \cos(w_2 t)$

ii. Show that $x_{usb}(t)$ can be demodulated by the synchronous demodulation

[Hint: Choose local oscillator carrier = $\cos(w_c t)$]

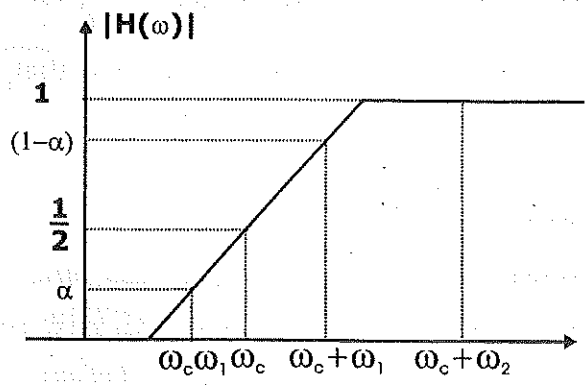

Fig 4(b)

5. a. A carrier is frequency modulated with a sinusoidal signal of 2KHz, resulting in a maximum frequency deviation of 5KHz. **06**

i. Find the band width of the modulated signal.

ii. The amplitude of the modulating sinusoid is increased by a factor of 3. And its frequency is lowered to 1KHz. Find the maximum frequency deviation and the bandwidth of the new modulated signal.

b. Derive time-domain expression for a wide band frequency modulating wave and show that **08**

$$S_{WBFM}(t) = Ac \sum_{n=-\infty}^{+\infty} Jn(\beta) \cos((W_c + nw_m)t)$$

6. a. For a WBFM if narrow band carrier f_1=0.1MHz, second carrier, f_2=9.5MHz, output carrier frequency f_c=100MHz and Δf=75KHz. Calculate multiplying factors n_1 and n_2. If NBFM frequency deviation is 20Hz. Draw the Block diagram of the modulator. **06**

b. Explain demodulation of FM using Balanced slope detector. Also comment on advantage and disadvantage. **08**

7. a. Derive an expression for overalls equivalent noise temperature and overall equivalent noise figure of the cascade connection of any number of noises for two part network. **06**

b. Derive the figure of merit of AM receiver and show that it is equal to **08**

$$FOM = \gamma = \frac{\mu^2}{2 + \mu^2}$$

8. a. Explain different pulse modulation techniques in analog domain. **06**

b. Calculate the transmission Band width B_T and the required transmission power S_T of DSB, SSB & AM system for transmitting an audio signal which has bandwidth of 10KHz with an output SNR of 40dB. Assume that the channel introduces a 40dB power loss and channel noise in AWGN with power spectral density, **08**

$$\frac{n}{2} = 10^{-9} W / Hz, \ \& \ \mu = 1 .$$